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Each feature must be completely safe

Samsung SRI-Noida MD on pushing more AI onto devices without compromise

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Samsung's India R&D story has long moved beyond cost arbitrage. Its Noida facility is now a critical node in the company's global innovation network, contributing significantly towards Artificial Intelligence, software-led differentiation and India-first problem solving that often scales globally. In a recent roundtable with select journalists, Kyungyun Roo, Managing Director of Samsung R&D Institute Noida (SRI-Noida), offered a detailed look at how the center is shaping the Galaxy S26 series and more importantly, how India is influencing Samsung's global product roadmap.

Kyungyun Roo
Managing Director of Samsung R&D Institute Noida (SRI-Noida)

Responding to the question on challenges of pushing AI capabilities onto devices, Roo summed up the central tension succinctly: the need to strike a balance between on-device processing and cloud capabilities.

"We all know about on-device and cloud-based AI elements," Roo said. "On-device is good for security and latency, but there are limitations. The important thing is balancing the on-device and the cloud side."

This balance is at the heart of Samsung's Galaxy AI strategy. While on-device AI ensures faster responses and stronger privacy protections, it comes with constraints in terms of computational power and model size. Cloud AI, on the other hand, offers scale but introduces latency and dependency on connectivity.

For the Galaxy S26 series, Samsung has leaned more heavily into its on-device AI model, Gauss. Compared to the Galaxy S24, Roo revealed, a significantly larger number of features now run locally on the device. This has been enabled by improvements in model efficiency and hardware capabilities.

"It's a continuous process," he explained. "Some features that were not possible earlier on-device are now possible because we've solved those challenges over time."

India's growing influence on global features

One of the most notable aspects of Samsung's R&D strategy in India is how local insights are increasingly shaping global features.

A standout example is Direct Voicemail, a feature that originated from Indian user feedback via the Samsung Members community. Initially launched in the Galaxy A series, it received strong traction and was eventually scaled up to the flagship Galaxy S26 lineup globally.

Similarly, the concept of backup calling, designed to address India's patchy network conditions and widespread dual-SIM usage, was conceived locally before it slowly became a standard feature across Samsung devices worldwide.

"In India, you may have two SIM cards because coverage varies," Roo noted. "The idea was: if one network fails, the other should seamlessly take over. This was proposed from India and adopted globally."

Designing for India means tackling a unique set of challenges: extreme temperatures, inconsistent network connectivity, linguistic diversity, and high-density usage environments. Roo pointed out that thermal performance is a major area of focus.

"India has very high temperatures, and device specifications are also increasing. We focus on performance optimisation to ensure stability across such conditions," he said.

Network instability is another area where SRI-Noida has contributed significantly. The team has developed AI-driven solutions to mitigate chunks of audio packet loss during calls by dynamically adjusting buffering based on historical data patterns.

Even features like Privacy Display have been shaped by India-specific behaviors. With OTPs (one-time passwords) widely used across banking and digital services, the feature intelligently masks sensitive information without disrupting usability.

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